Real-Time Intrinsic TND Valuation Model

IN 22-21 — Times Series Analysis

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Class : A

1. Introduction

This report presents a real-time intrinsic value model for the Tunisian Dinar (TND) against the US Dollar (USD). The model combines two components: a basket-based fundamental value derived from global currency movements, and a stochastic adjustment for local Tunisian market conditions. Together, these produce a continuously updating fair-value estimate for USD/TND that accounts for both international fundamentals and local liquidity pressures.

The model addresses a key limitation of the official BCT fixing rate , it is published only twice daily and cannot capture intraday market movements. The intrinsic value model fills this gap by providing a real-time estimate anchored to live international FX data and adjusted for the historically observed gap between the interbank rate and the fixing.

2. Model Framework Design

The intrinsic value model is built on two components that combine to produce a final real-time estimate:

2.1 Component 1 : Basket-Based Baseline

The baseline captures the fundamental value of USD/TND as implied by movements in three major international currency pairs: EUR/USD, GBP/USD, and USD/JPY. The BCT fixing is known to respond to these global currency movements, as Tunisia's economy is heavily integrated with Europe and global dollar dynamics.

The baseline formula is:

$$\text{Baseline}(t) = \text{FIX}(t-1) \times (1 + w_1 \times \%\Delta\text{EUR/USD} + w_2 \times \%\Delta\text{GBP/USD} + w_3 \times \%\Delta\text{USD/JPY})$$

where FIX(t-1) is yesterday's BCT fixing mid-rate and the weights w_1, w_2, w_3 are estimated via OLS regression. Each day the baseline is updated as international markets move, providing a real-time anchor.

2.2 Component 2 : Stochastic Liquidity Adjustment

The second component captures the deviation caused by local Tunisian FX market conditions. In practice, banks trade USD/TND at the interbank (IB) rate, which differs from the BCT fixing due to local liquidity pressures, supply-demand imbalances, and market frictions.

This deviation — the spread between the IB rate and the BCT fixing — is modeled as an AR(1) process:

$$\text{Spread}(t) = \alpha + \varphi \times \text{Spread}(t-1) + \varepsilon(t)$$

where α is the intercept, φ is the autoregressive coefficient, and $\varepsilon(t)$ is white noise. Since the official IB rate is published with a one-day lag, the AR(1) model nowcasts today's spread using yesterday's observed value.

2.3 Final Intrinsic Value

The two components are combined additively to produce the final intrinsic value:

$$\text{Intrinsic Value}(t) = \text{Baseline}(t) + \text{Nowcasted Spread}(t)$$

This output represents the best real-time estimate of where USD/TND should be trading, accounting for both global currency fundamentals and current local market conditions.

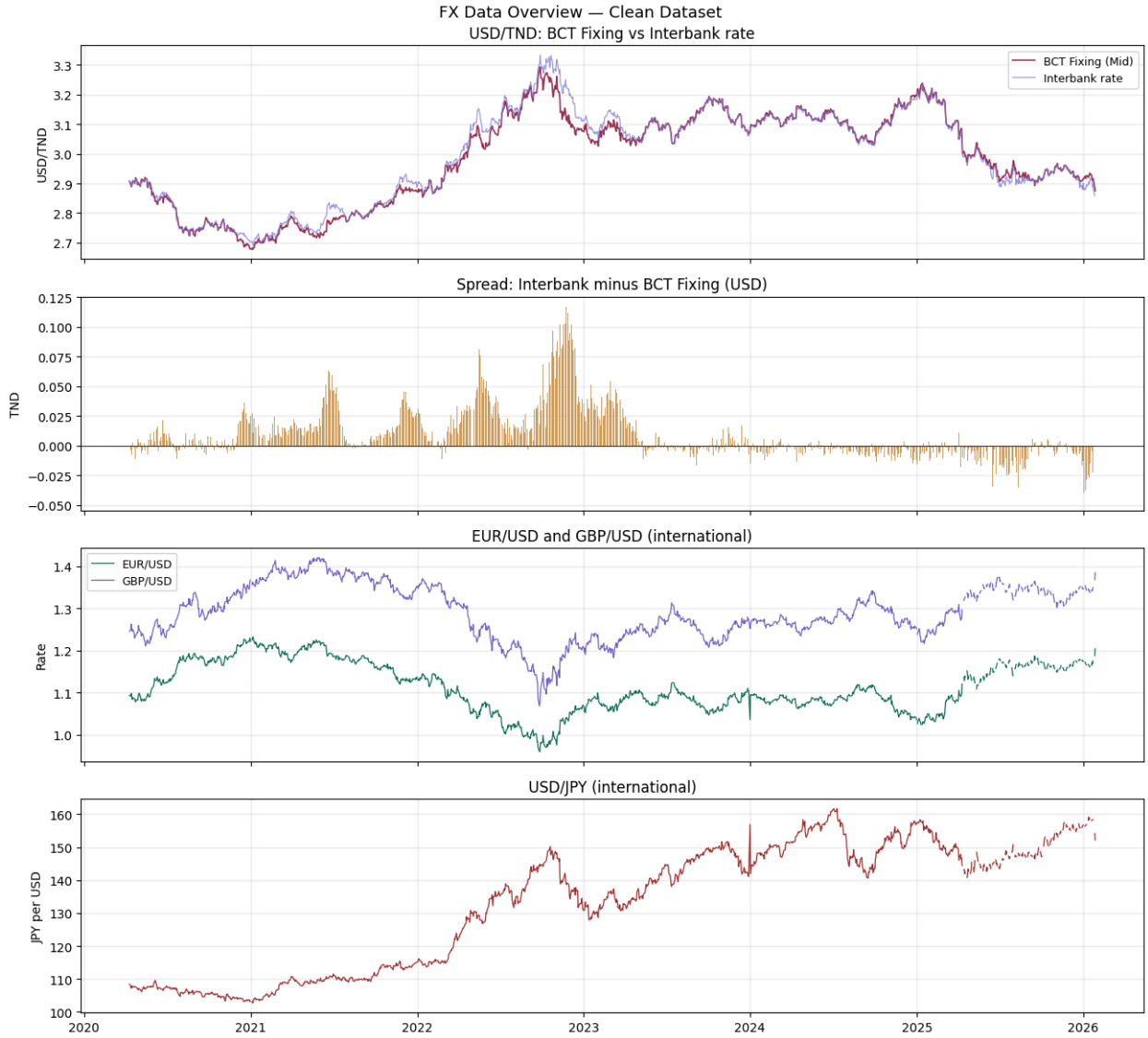
3. Data

All data was sourced from a single Excel file containing four sheets covering the period April 2020 to January 2026:

- InterBancaire: daily interbank USD/TND, EUR/TND, GBP/TND, and JPY/TND rates (2004–2026)
- BCT Fixing USD: official BCT fixing for USD/TND Bid, Ask, and Mid (2020–2026)
- BCT Fixing EUR: official BCT fixing for EUR/TND Bid, Ask, and Mid (2020–2026)

- EUR-GBP-JPY vs USD: international EUR/USD, GBP/USD, and USD/JPY closing rates (2020–2026)

After cleaning, aligning on a common date index, and removing weekends and incomplete rows, the final dataset contains 1,513 trading days. The Mid price was used as the reference fixing rate throughout, as it represents the fair midpoint between the Bid and Ask prices.



4. Basket Weight Estimation

An Ordinary Least Squares (OLS) regression was estimated to quantify the relationship between daily log returns of EUR/USD, GBP/USD, and USD/JPY and the daily log

return of the BCT USD/TND fixing mid-rate. The regression was run on 1,342 complete observations over the full sample period.

4.1 Regression Results

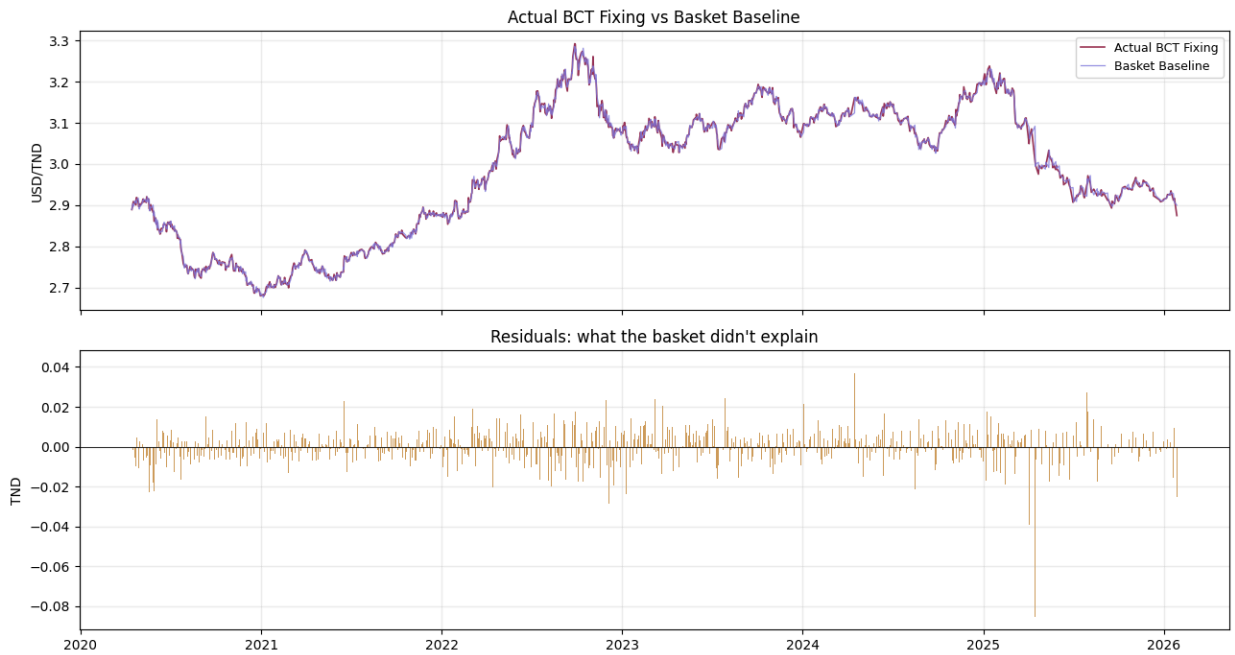
Variable	Coefficient	Std Error	t-statistic	p-value
Constant	0.0000339	0.0000685	0.495	0.621
EUR/USD (w_1)	−0.3520	0.023	−15.494	0.000
GBP/USD (w_2)	−0.0318	0.019	−1.663	0.097
USD/JPY (w_3)	0.0457	0.013	3.415	0.001

Metric	Value
R-squared	0.363
Adjusted R-squared	0.361
F-statistic	253.9
Observations	1,342

4.2 Interpretation

EUR/USD is the dominant driver of the TND fixing with a coefficient of −0.352, statistically significant at the 1% level. The negative sign reflects the inverse relationship: when the Euro strengthens against the dollar internationally, the dollar weakens and USD/TND decreases. This is economically intuitive given Tunisia's deep trade ties with the Eurozone.

USD/JPY carries a small positive weight of +0.046, also statistically significant. GBP/USD has a weight of −0.032 with a p-value of 0.097, indicating a weaker and less reliable relationship. The model explains 36.3% of daily fixing movements — a reasonable result given that the BCT smooths its fixing and does not mechanically follow international markets every day.

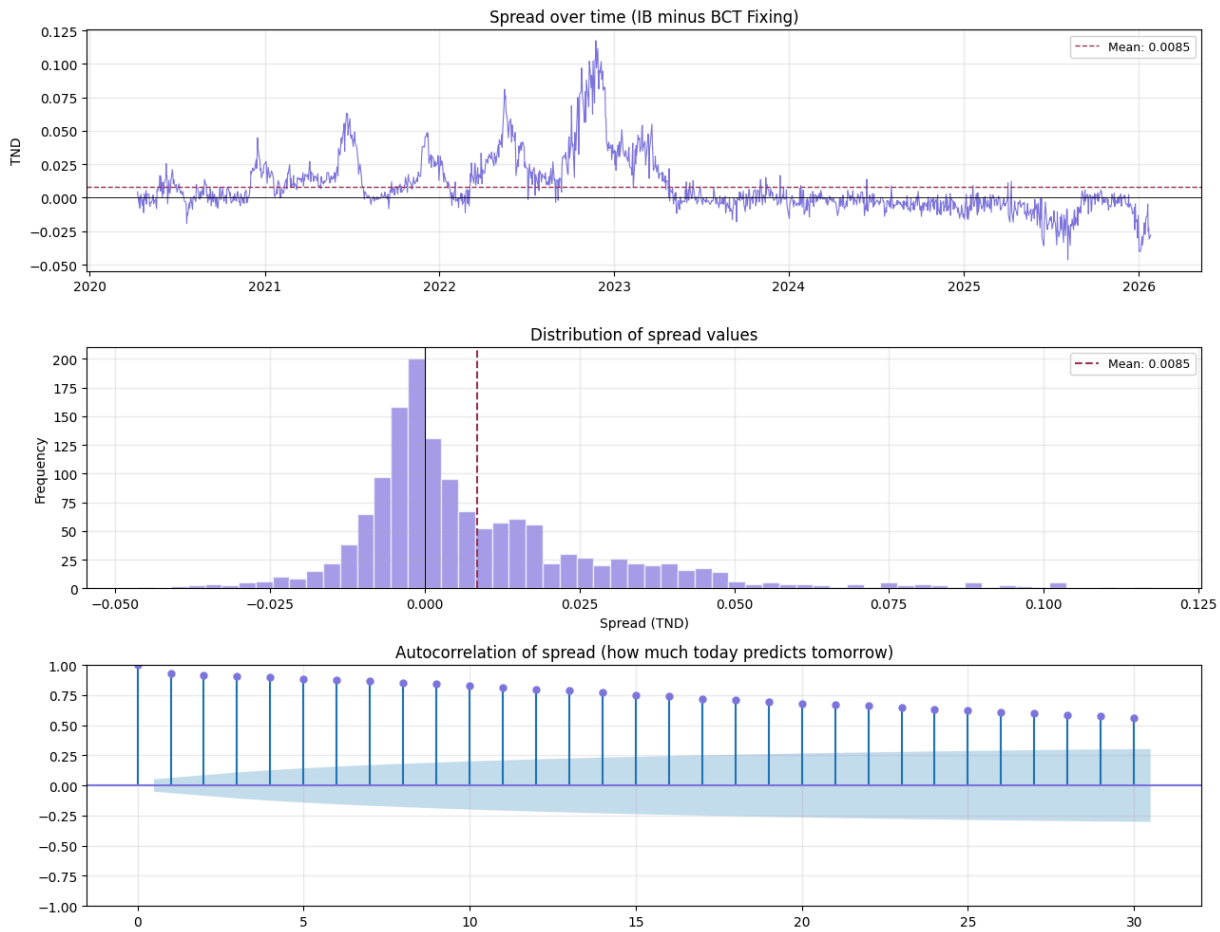


5. Interbank Rate Nowcasting

5.1 Spread Analysis

The spread between the interbank rate and the BCT fixing was analyzed over the full sample period. Key statistics are presented below:

Statistic	USD Spread
Mean	+0.0085 TND
Std Dev	0.0214 TND
Min	−0.0465 TND
Max	+0.1173 TND
Lag-1 Autocorrelation	0.930
ADF p-value	0.051



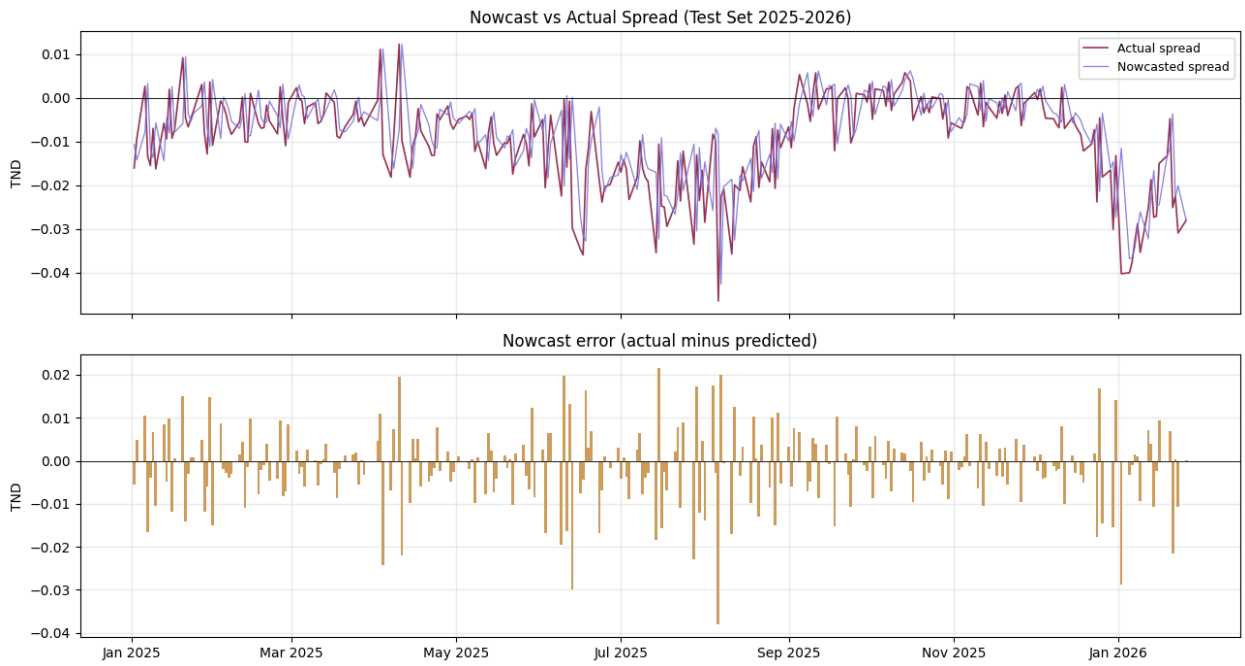
Three distinct regimes were identified. During 2020–2021 the spread was calm and near zero. The 2022–2023 period saw a severe dollar scarcity crisis with the spread peaking at 0.117 TND, reflecting extreme liquidity pressure in the Tunisian FX market. From 2024 onwards the spread turned negative, indicating a shift to dollar surplus conditions.

The lag-1 autocorrelation of 0.930 confirms that the spread is highly persistent — today's spread is an extremely strong predictor of tomorrow's. This persistence justifies the use of an AR(1) model for nowcasting.

5.2 AR(1) Nowcasting Model

An AR(1) model was fitted on training data covering 2020–2024 (1,196 observations). The estimated model is:

$$\text{Spread}(t) = 0.000807 + 0.9340 \times \text{Spread}(t-1)$$



The autoregressive coefficient of 0.934 confirms extremely high persistence. The intercept of 0.000807 implies a long-run average spread slightly above zero, consistent with the observed data. A Kalman filter was considered but not adopted — given the very high autocorrelation, the AR(1) model captures the predictable structure of the spread with sufficient accuracy.

The one-step-ahead nowcasting approach uses only information available at the start of each trading day: yesterday's actual IB rate. This makes the model operationally realistic despite the official one-day publication lag.

6. Performance Evaluation & Backtesting

6.1 Test Set Performance (2025–2026)

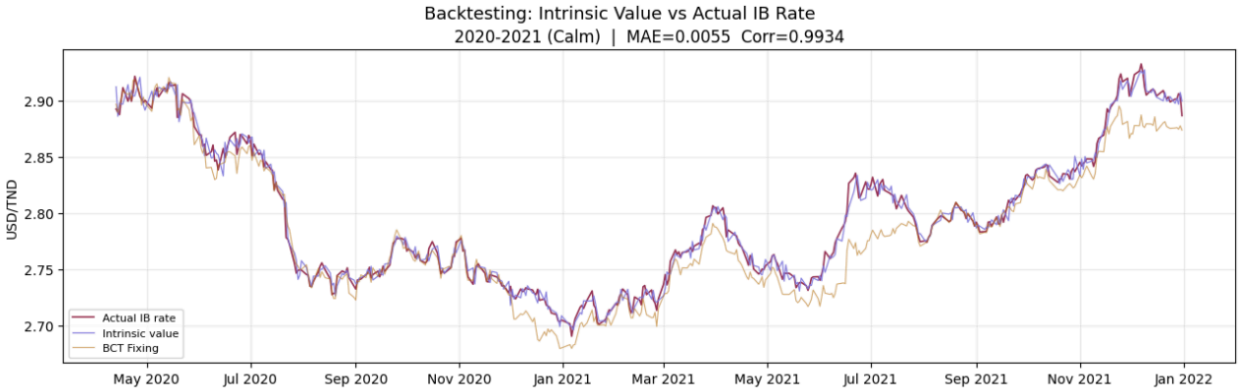
The model was evaluated on an out-of-sample test period covering 2025 to January 2026 (265 observations). Performance was measured against the actual published IB rate:

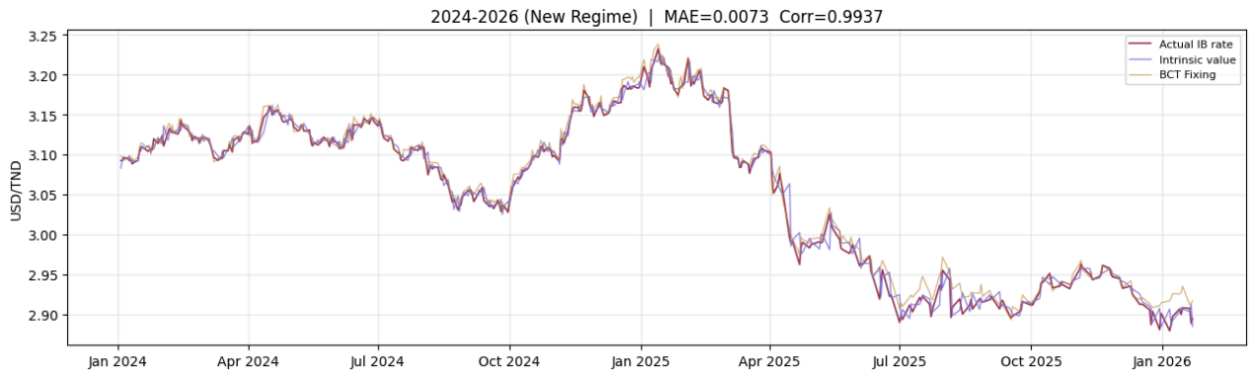
Metric	Nowcast Spread	Intrinsic Value vs IB Rate
MAE	0.00647 TND	0.00952 TND
RMSE	0.00879 TND	0.01313 TND
Correlation	0.613	0.993

The intrinsic value tracked the actual IB rate with a correlation of 0.993 and a mean absolute error of under 1 fil less than 0.3% of the prevailing exchange rate level. The spread nowcast achieved a correlation of 0.613, indicating that while daily directional prediction is imperfect, the level of the spread is well captured.

6.2 Backtesting Across Market Regimes

Period	Regime	Observations	MAE (TND)	RMSE (TND)	Correlation
2020–2021	Calm	435	0.00549	0.00711	0.9934
2022–2023	Crisis	507	0.00846	0.01115	0.9936
2024–2026	New Regime	432	0.00727	0.01045	0.9937





The model demonstrates robust performance across all three market regimes. Correlation exceeds 0.993 in every period including during the 2022–2023 crisis when spread volatility was at its highest. MAE is naturally higher during the crisis period (0.00846 TND) compared to calm conditions (0.00549 TND), reflecting the greater difficulty of predicting extreme spread spikes.

A key finding from the backtesting is that during the 2022–2023 crisis the intrinsic value tracked the actual IB rate more closely than the official BCT fixing. The fixing lagged behind the market by a significant margin while the intrinsic value anchored to both global fundamentals and the nowcasted spread captured the market movement in real time. This validates the core objective of the model.

7. Limitations & Extensions

- The basket regression explains only 36% of daily fixing movements. The remaining 64% reflects BCT policy decisions that are not mechanically tied to international markets and cannot be predicted from global FX data alone.
- The AR(1) spread nowcast has moderate directional accuracy (correlation = 0.61). On days when the spread makes sudden large moves typically due to unexpected policy shifts or external shocks the model will lag by one day.
- The model assumes a stable relationship between international FX pairs and the TND fixing over time. Structural breaks such as a change in BCT policy framework would require re-estimation of the basket weights.
- A natural extension would be to replace the AR(1) with a Kalman filter, which could incorporate intraday signals such as live indicative quotes from major banks to update the spread estimate continuously throughout the day.
- Including additional explanatory variables such as oil prices, Tunisia's trade balance, or foreign reserve levels could improve the basket regression's explanatory power.

8. Conclusion

This project successfully delivered a real-time intrinsic value model for USD/TND. The two-component framework , combining a basket-based fundamental baseline with an AR(1) nowcast of the IB-fixing spread — produces a continuously updating fair-value estimate that tracks the actual interbank rate with over 99% correlation across all tested market conditions.

The model demonstrates that even a parsimonious approach, using publicly available international FX data and a simple autoregressive model for the local spread, can provide meaningful real-time insight into TND valuation beyond what the official BCT fixing offers. The results are particularly compelling during the 2022–2023 dollar scarcity crisis, where the intrinsic value correctly identified market stress that the fixing rate was slow to reflect.